

"Falcon 2.0T: System voltage low from failed 180A alternator"

Document: tsb-20-511

Area: bulletins

1 Condition

Some 2.0T Falcon vehicles may exhibit a battery warning lamp, dimming lights, or a no-start condition accompanied by DTC P0562 (System Voltage Low) and DTC P0620 (Generator Control Circuit). Drivers may report stalling under high electrical load or repeated dead-battery complaints. Both codes may be stored together when charging output falls below the regulated target. See Falcon Charging Service Manual :: System Voltage Codes for the conditions that set P0562.

2 Cause

DTC P0562 and DTC P0620 on these vehicles are most often caused by a failed 180A alternator ALT-8810 with worn brushes or an internal regulator fault that interrupts the generator control circuit. A loose or corroded B+ output connection can produce the same low-voltage symptoms. Review charging system operation in Falcon Charging Service Manual :: Alternator and the control circuit description in Falcon Charging Service Manual :: Generator Control Circuit Code before condemning ALT-8810.

3 Diagnosis

Perform charging system test PROC-CHG-DIAG to measure regulated output voltage and ripple under load before replacing the alternator. Confirm battery state of charge, inspect the B+ cable and ground, and verify the generator control circuit communicates correctly with the ECM. If output remains below specification with P0562 and P0620 current, the alternator ALT-8810 is the confirmed failure. The full test sequence appears in Falcon Charging Service Manual :: Charging System Diagnosis Procedure .

4 Repair Procedure

Replace the 180A alternator ALT-8810 following procedure PROC-ALT, taking care to disconnect the battery before servicing the B+ terminal. After installation, verify regulated charging voltage and confirm DTC P0562 and P0620 do not return after a road test. Inspect and retorque the output cable to specification. Detailed removal and installation steps are provided in Falcon Charging Service Manual :: Alternator Replacement Procedure , with supporting overview information in Falcon Charging Service Manual :: Charging System Overview .